AI and Robotics

Lesson 1a: Introduction

Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

https://youtu.be/cRZNwgvcWUg?si=kIfNi9zTUdHsXraJ&t=787

”Pick up an apple”

”Pick up an apple”

”I am hungry”

AI and Robotics

Sarah Keren

Objectives

History of Robotics

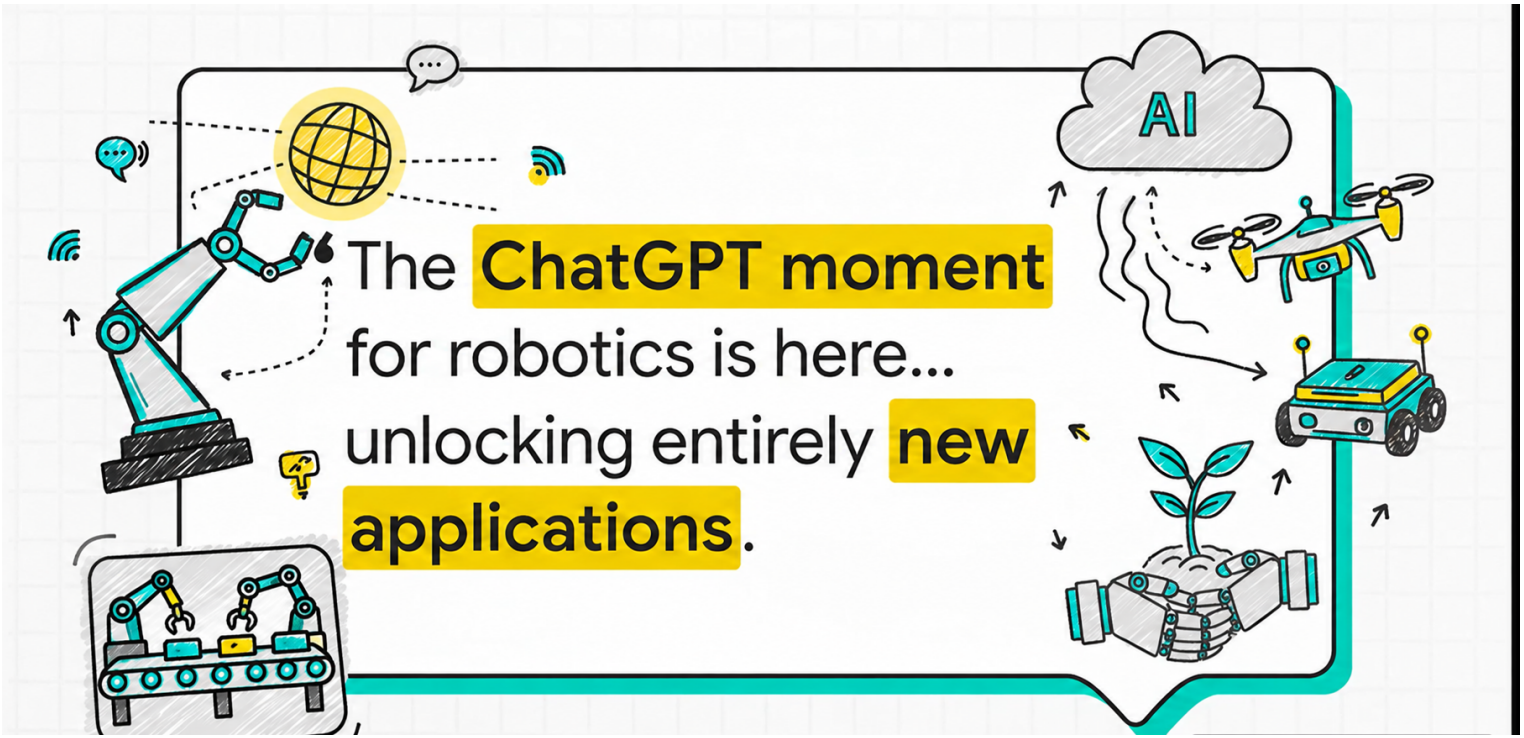
Studying Robotics

CLAIR lab and Team

Our Running Example

Course Overview

Objectives



AI and Robotics
Sarah Keren
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

AI and Robotics

AI develops methods that enable agents to perceive, reason, learn, and act to achieve goals.

Robotics studies the design and operation of embodied systems that sense and act in the physical world.

At their intersection, AI enables robots to interpret observations, plan actions, and adapt to uncertain and changing environments.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Physical AI

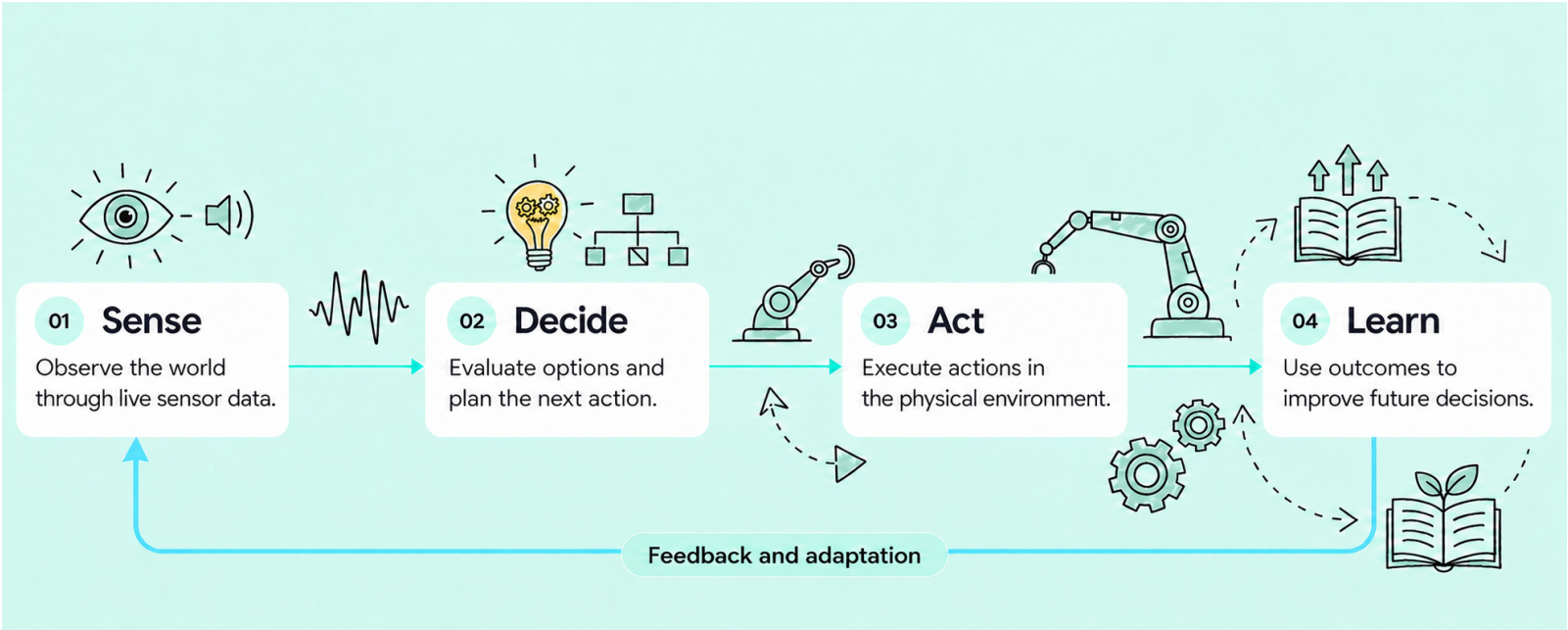
Unlike traditional abstract AI, **Physical AI** must account for the properties and constraints of the physical environment.

Currently a ‘hot’ topic, but underlying research builds on decades of work in robotics, embodied intelligence, planning, control, reinforcement learning, and autonomous systems.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

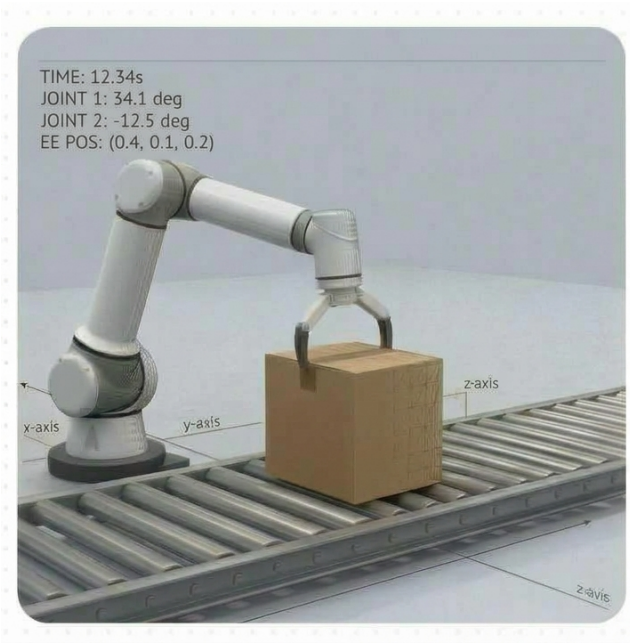
The Continuous Closed Loop

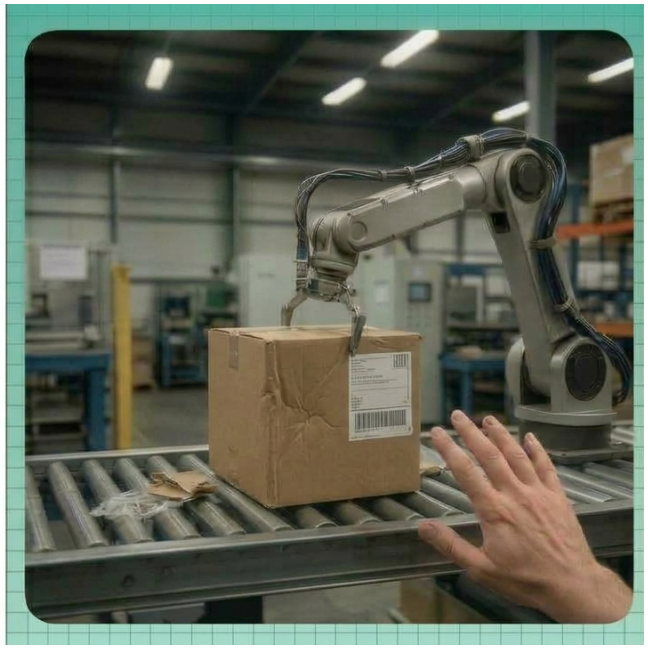
Continuous interaction between intelligence and the physical world.



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Sim-to-Real Gap

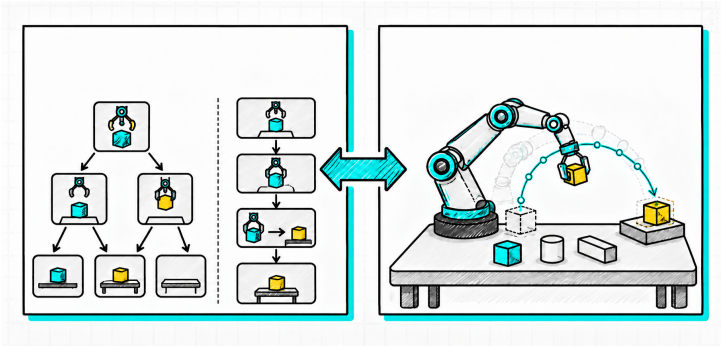




Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Our Objectives

Provide a practical understanding of how AI-drive robotic systems combine **discrete task decisions** with **continuous** geometry, collision **constraints**, and **feasible** robot motion.



Account for **uncertainty and imperfect** sensing, changing environments, and execution **failures**.

Implement and evaluate core algorithms in simulation and on physical robotic platforms.

Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

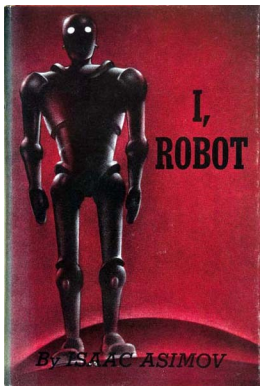
Our Running Example

Course Overview

History of Robotics

What is a robot ?

The word 'Robot' was coined by a Czech playwright named Karel Capek in his 1921 play "Rossum's Universal Robots"
Rabota = menial work
The word "Robotics" believed to be coined by Isaac Asimov

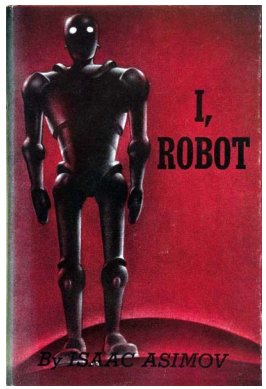


Definitions?

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

What is a robot ?

The word 'Robot' was coined by a Czech playwright named Karel Capek in his 1921 play "Rossum's Universal Robots"
Rabota = menial work
The word "Robotics" believed to be coined by Isaac Asimov



Definitions?

- **Definition:** A **robot** is an autonomous system that exists in the **physical** world, can sense its environment, and can act on it to achieve some goals.
- **Definition:** An **autonomous robot** acts based upon its decisions, and is not controlled by a human.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

A Brief History of Robotics

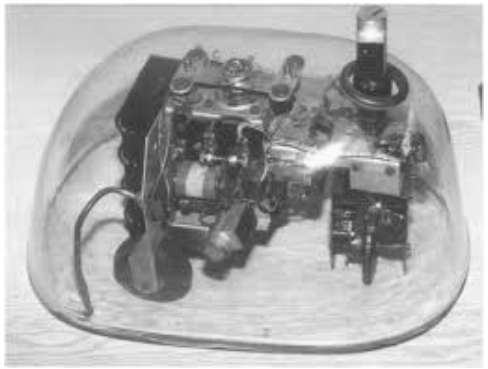
Robotics emerged from the influences of:

- **Control Theory:** the mathematics of controlling machines.
  - Develop algorithms governing to drive the system to a desired state, while minimizing delay, maximizing stability, etc.
  - In robotics,feedback control used sensors and making adjustments to keep the measured variable within a set range
- **Cybernetics:** the study of control and communication in biological systems and machines
  - Integration of sensing, action, and the environment
- **Artificial Intelligence:** the mechanisms for planning and reasoning;
  - The study of intelligent agents, which refers to any system that perceives its environment and takes actions that maximize its chance of achieving its goals
  - The original focus was on creating machines that think like humans (i.e., that pass the Turing test) - today the focus has changed.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

The Tortoise Robot

- Grey Walter (1910-1977) - a neurophysiologist and cyberneticist who created in the 1940s simple robots based on reflexes (before AI existed) - bio-mimetic machines.
- Behaviors:
  - find light / head towards light / back away from light
  - turn and push to avoid obsacles recharge battery
- Control: reactive, with prioritized reflexes
- First example of 'artificial life': Tortoises now shown in museums around the world



https://youtu.be/lLULRlmXkKo

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Artificial Intelligence

- Originate in 1956 at a workshop at Dartmouth University, USA: **Dartmouth Summer Research Project on Artificial Intelligence**
- Some of the researchers present: Marvin Minsky, John McCarthy, Allan Newell, Herbert Simon, Claude Shannon, John Nash, etc..

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Artificial Intelligence

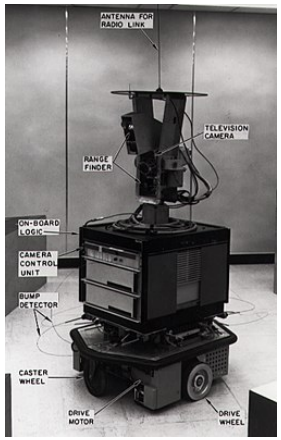
- Outcome of meeting - what we need for AI is:
  - Internal models of the world
  - Search through possible solutions
  - Planning and reasoning to solve problems
  - Symbolic representations of information
  - Hierarchical system organization
  - Sequential program execution

” We propose that a 2-month, 10-man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College. ”

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Shakey: the First intelligent robot

- Built in the 1960s at the Stanford Research Institute (SRI)
- The robot had
  - Camera and contact sensors
  - Application of AI techniques to vision algorithms
  - Lived in a simple black and white world
  - The robot shook when trying to execute its navigation plans.
- Reasoning:
  - Planning with STRIPS
  - Programming with LISP

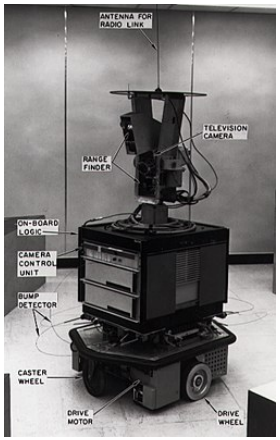


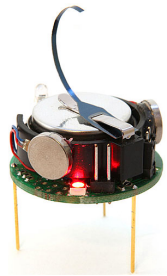
https://www.youtube.com/watch?v=7bsEN8mwUB8&t=21s

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

A New Paradigm: Behavior Based Robotics

- Robots developed based on classical AI techniques in the 1970s and 1980s thought to be **too hard** and **too slow**.
- 1980s robotics enters a new phase:
  - reactive control
  - hybrid control
  - behavior-based control
- In stark contrast to purely **deliberative** control





Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Behavior Based Robotics

A new wave in AI in the 80s-90s

- Spearheaded by Rodney Brooks (MIT) and later Ronald Akin (Georgia Tech)
- Stepping away from complex AI, deliberative planning (e.g., a chess playing computer) to the basics of intelligence.
- How? By realizing that organisms in nature follow simple sets of **reactive rules** giving rise to **reactive behaviors**

Which behavior-based robot do we all know?

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Behavior Based Robotics

A new wave in AI in the 80s-90s

- Spearheaded by Rodney Brooks (MIT) and later Ronald Akin (Georgia Tech)
- Stepping away from complex AI, deliberative planning (e.g., a chess playing computer) to the basics of intelligence.
- How? By realizing that organisms in nature follow simple sets of **reactive rules** giving rise to **reactive behaviors**

Which behavior-based robot do we all know?

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Behavior Based Robotics

Behavior-based architectures had a huge impact on collective intelligence and swarm robotics

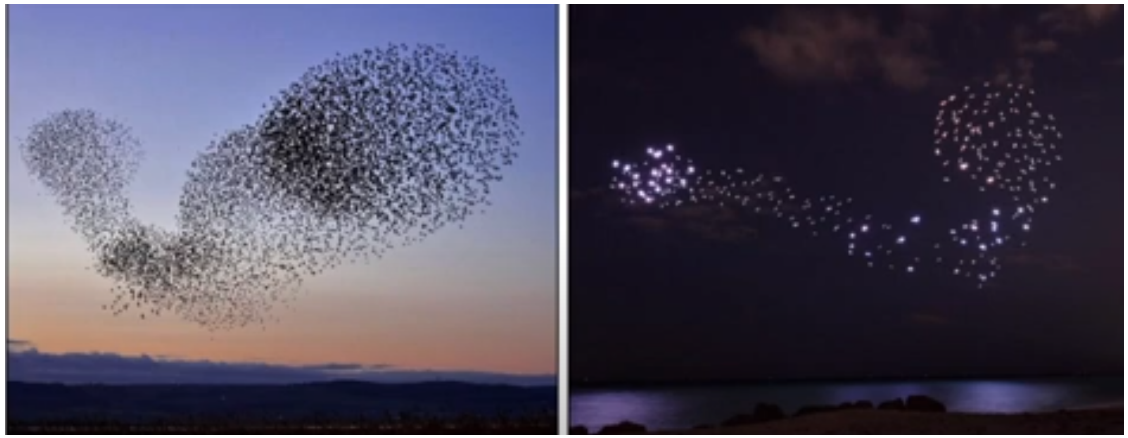


Figure 1: Starlings

`https://youtu.be/0bRocfcPhHU?t=255`

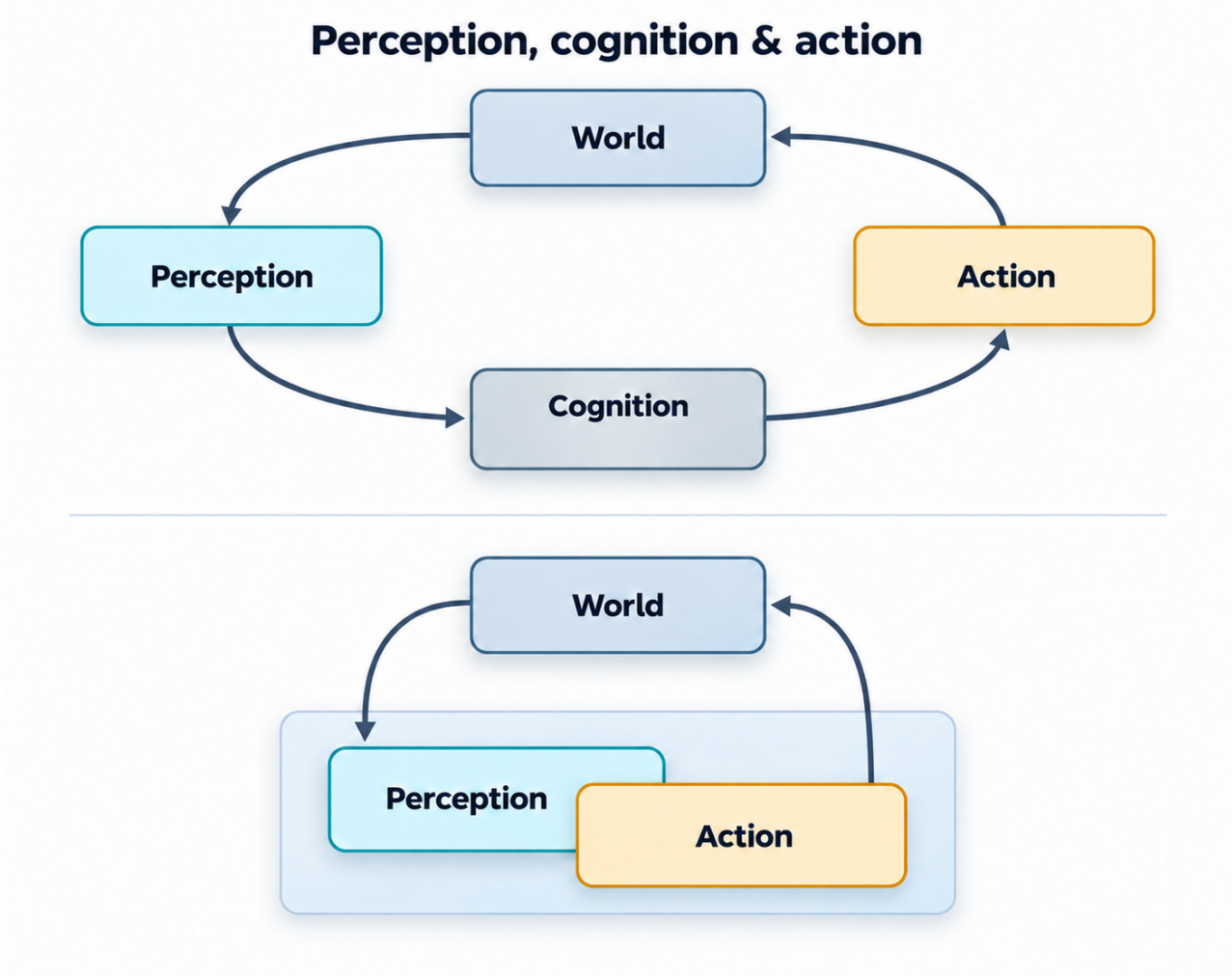
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Cognitive robotics

- A subfield of robotics concerned providing a robot with a processing architecture that will allow it to learn and reason about how to behave in response to complex goals in a complex world.
- Needs to
  - monitor itself for potential problems
  - schedule tasks in time
  - come up with novel plans to achieve desired goals over time
  - collaborate with other (autonomous) agents
  - deal with risk ...

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Two streams of thought in robotics



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Studying Robotics

Why Study Robotics ?

Figure 2: Yesterday's robots

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Why Study Robotics ?

Figure 3: Way in the future robots

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Why Study Robotics ?

Figure 4: Today's robots

Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

Our Running Example

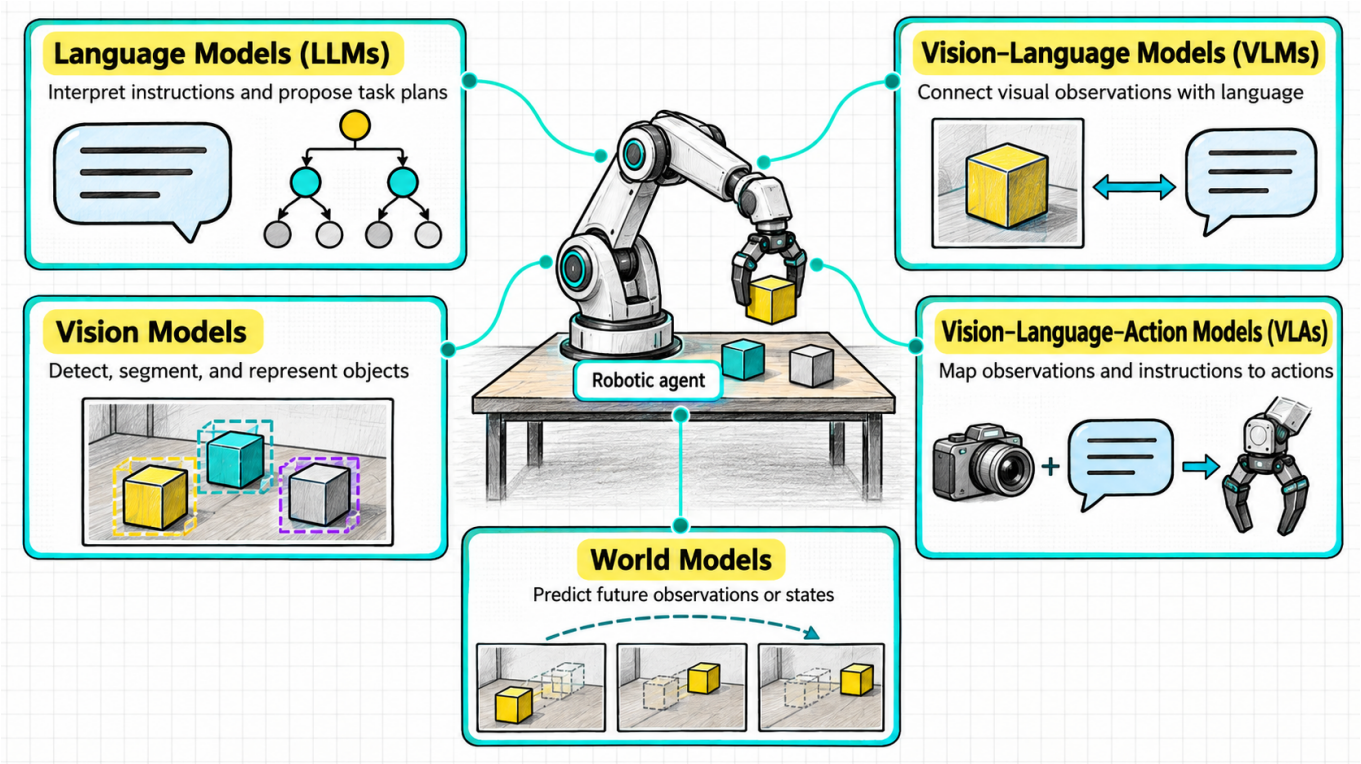
Course Overview

Why Study Robotics ?

Figure 5: Today's robots

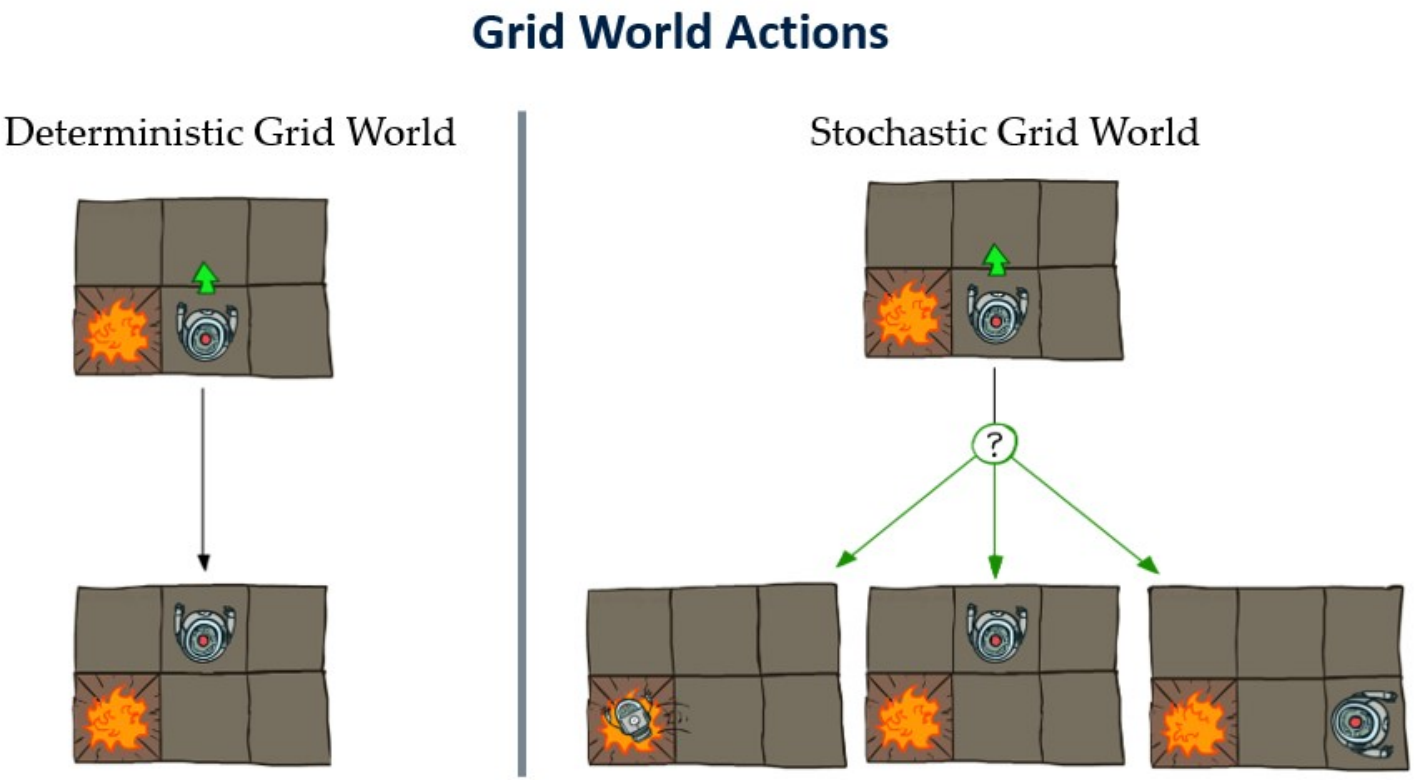
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Foundation Models for Robotics



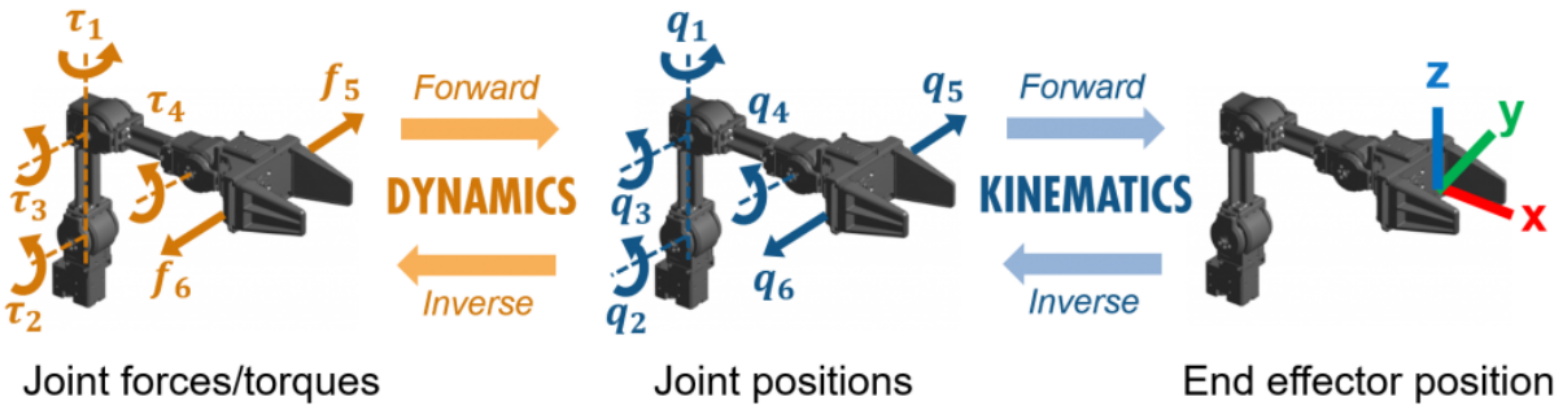
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Why take this course if you already took IAI?



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Why take this course if you already took a robotics course?



Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

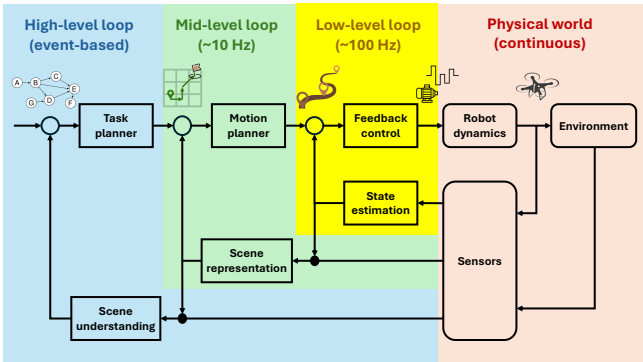
Our Running Example

Course Overview

Why take this course even if you are not really interested in robotics?

- It's not just about machines that move - it's about understanding how intelligent systems perceive, reason, and act in the real world.
- This requires algorithms to deal with uncertainty, noise, and delayed feedback.

Our focus is on designing **intelligent and adaptive [embodied] systems** that can make decisions and act effectively in complex environments.



History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Why take this course if foundation models can be used to solve robotics ?

NVIDIA's humanoid based on foundation models

History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Key Point:

Robots are quickly moving out of the 'factory' and into the 'real-world'. Soon there will be robots for the masses, but we aren't there yet...

Autonomy: Complex decision making in a complex world
Sensing, Thinking, Acting

A robot does not have to be a single physical object : Any combination of sensors and actuators can be thought of as a robot (smart home, factory, ...)

Autonomy

Challenges:

- How to model and perceive the world?
- How to process information and execute actions?
- How to reason and plan in the face of uncertainty?

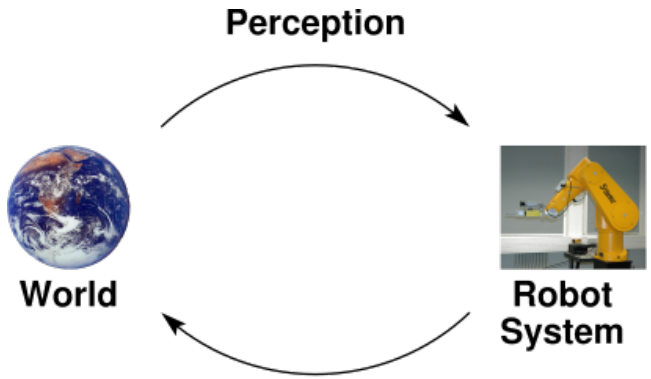
The field of robotics provides methods that address each challenge and architectures that combine them.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Autonomy

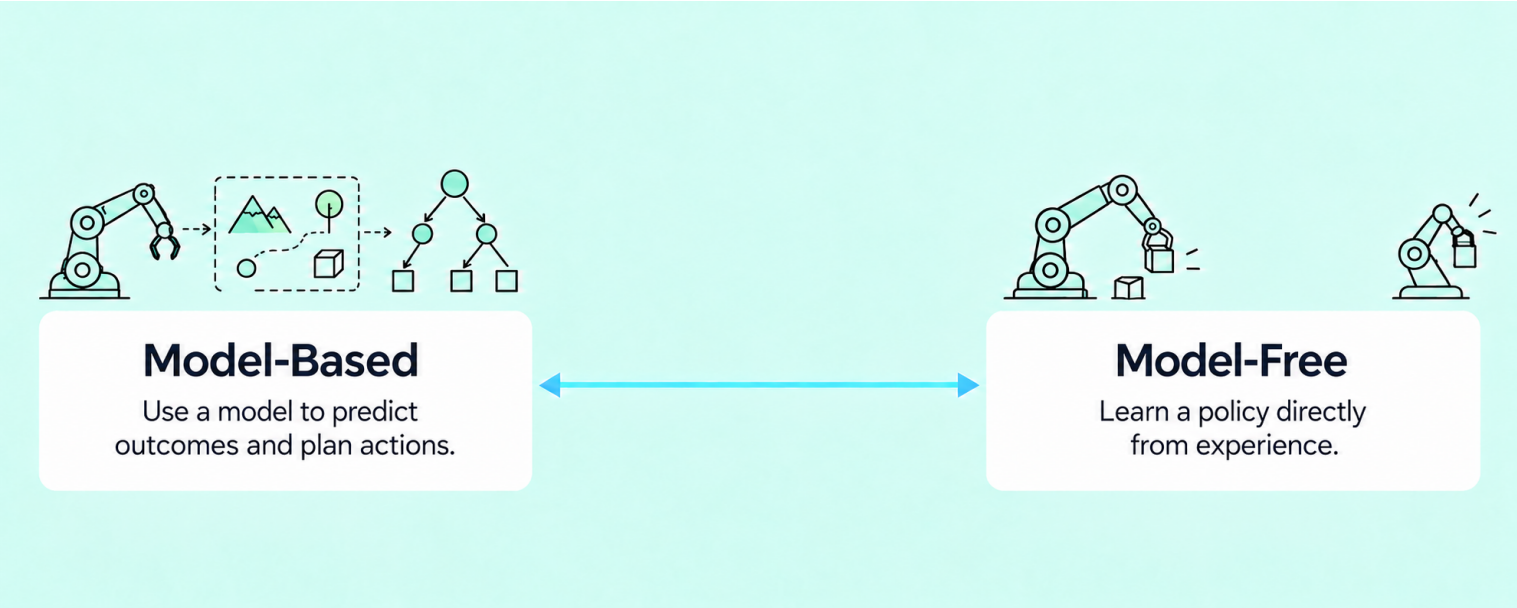
What does it mean to be autonomous ?

- **Action (Actuation)**
  - Locomotion: wheels (e.g., differential drives)
  - Manipulation: arms, grippers, etc.
- **Perception (Sensors)**
  - Proprioception (internal: IMU (inertial measurement unit), encoders , etc,)
  - Exteroception (external: cameras, scanners)
- **Cognition (Control)**
  - From reactive to proactive
  - From finite state machines to cognitive robotics



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Model-free vs. Model-based



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Model-free vs. Model-based

Model-based

- Has (or learns) a **model** of the world: $T(s' \mid s, a)$, $R(s, a)$
- Chooses actions by **planning** with the model (search, value iteration, MPC, MCTS)
- \+ Sample efficient, can reason about new goals
- – Model errors lead to bad plans (*model bias*)

Model-free

- No explicit model of the dynamics
- Learns a **policy** $\pi(a \mid s)$ or **value** $Q(s, a)$ directly from experience (Q-learning, policy gradient, PPO)
- \+ Simple, no modeling effort, strong asymptotic performance
- – Needs a lot of interaction data, hard to transfer to new tasks

Hybrids combine both: learn a model, then use it for planning or training

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Uncertainty

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Uncertainty

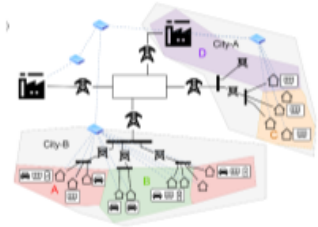
Sources of uncertainty:

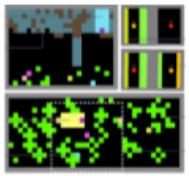
- **Perception** – noisy, partial sensor readings (*partial observability*)
- **Action** – actuators slip, grasps fail (*stochastic outcomes*)
- **Model** – dynamics and environment only approximately known

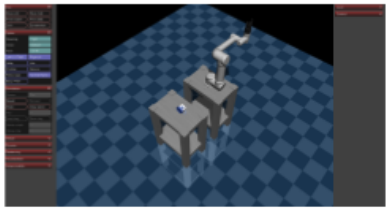
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

CLAIR lab and Team

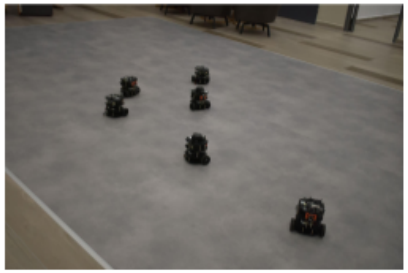
CLAIR







Collaborative AI and Robotics



TECHNION | The Henry and Marilyn Taub Faculty of Computer Science

46

Objectives

History of Robotics

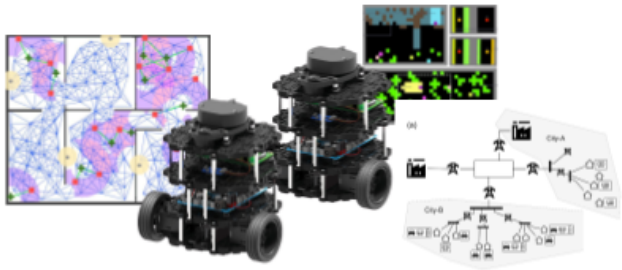
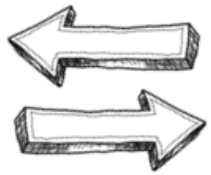
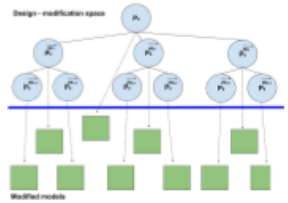
Studying Robotics

CLAIR lab and Team

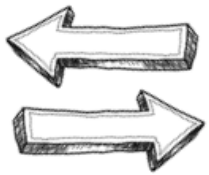
Our Running Example

Course Overview

CLAIR

Provide theoretical foundations for designing AI systems that are capable of effective partnership in sustainable and efficient collaborations of automated agents as well as of automated agents and people.

experience (data)  knowledge

47

Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

Our Running Example

Course Overview

Team: Sarah Keren

Head of the Collaborative AI and Robotics (CLAIR) lab

Developing novel approaches to identify AI agents' inherent capabilities and limitations and modify their environment to maximize performance and promote effective multi-agent collaboration.

Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

Our Running Example

Course Overview

Team: David Dovrat

- Roboticist
- **PhD Research:** An Automata Theory Method for the Analysis of Unicycle Pursuit Problems", supervised by Prof. Alfred M. Bruckstein, Technion.
- This week could not have happened without him.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Team: Guy Azran

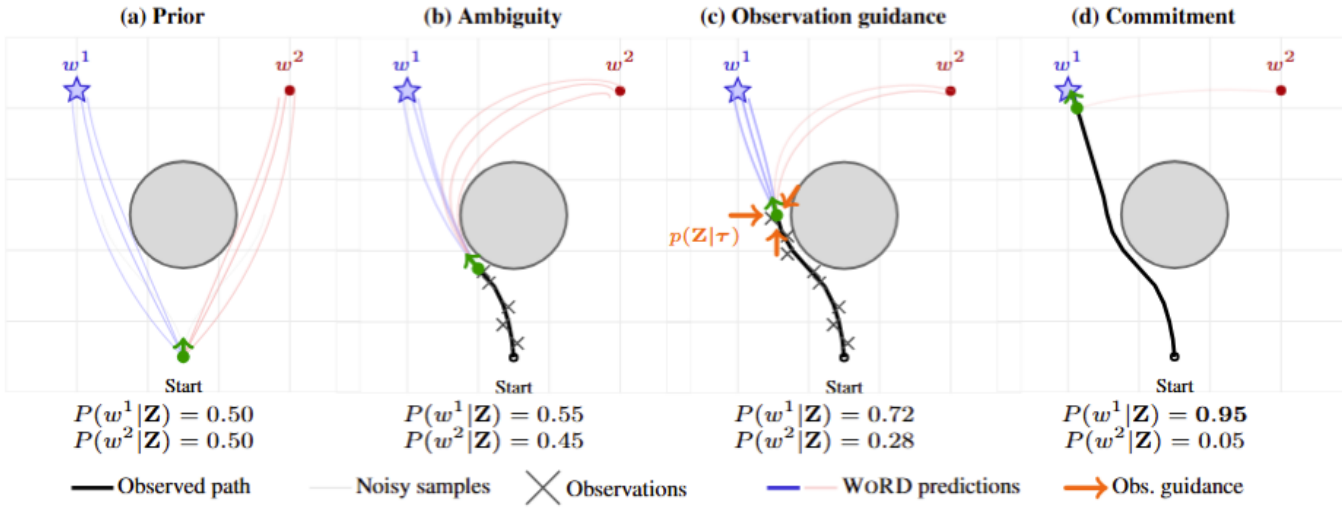
- PhD, advised by Sarah Keren
- **Research:** Robust task-aware state estimation
- Robust Task-Aware State Estimation from Pretrained (NeurIPS 2026)
- **This week:** Sunday-Tuesday tutorials

Figure 1: Grocery task from Example 1.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

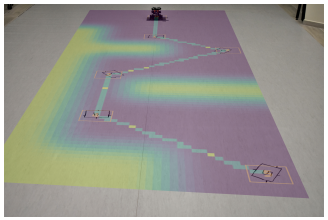
Team: Almog Anschel

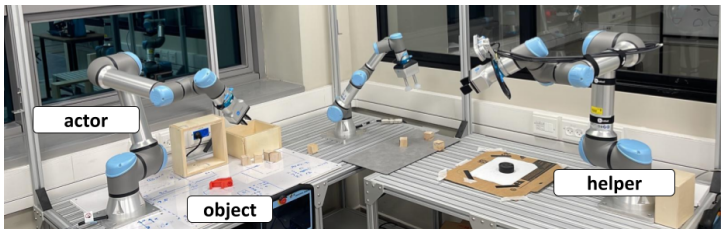
- MSc (finishing) → PhD, advised by Sarah Keren
- **Research:** diffusion models for robot planning and goal recognition
- **WoRD**: Waypoint Recognition with Diffusion (NeurIPS 2026)
- **This week:** Wed tutorial: Intro to Robotics with Deep Learning and World Models

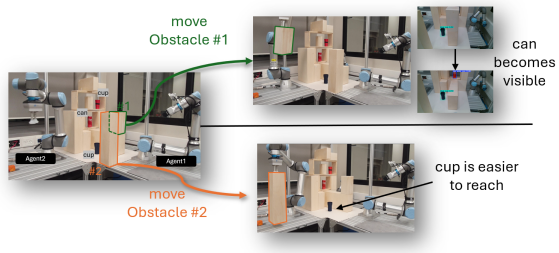


Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Work done at the CLAIR lab







Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Our Running Example

Example

The person: ”pick up an apple”

What do we need ?

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Perception and Scene Understanding

Sense and interpret the environment to locate an apple and estimate its properties.

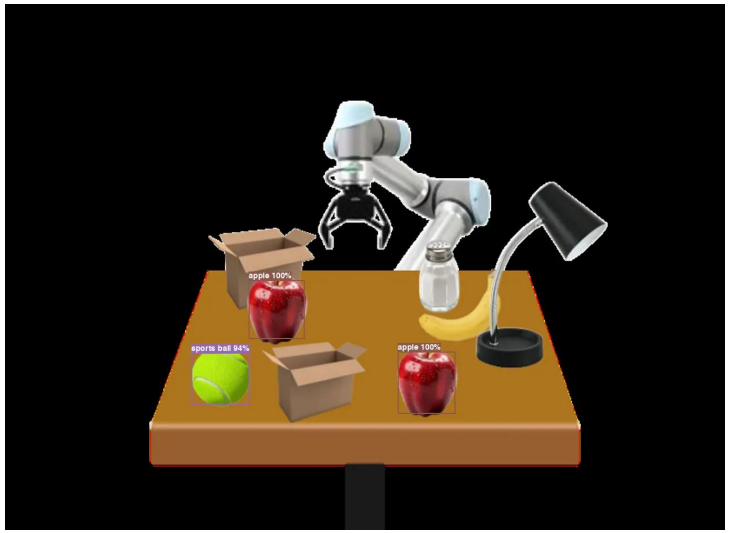
Sensor Data Acquisition: for example, RGB-D camera (color + depth), LiDAR, stereo camera, tactile and proximity sensors...

`https://huggingface.co/spaces/nielsr/dpt-depth-estimation`

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Perception and Scene Understanding

detect and classify objects in the image

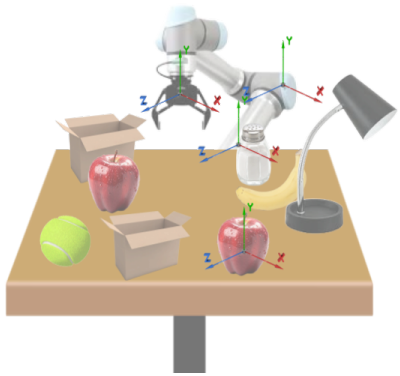


YOLO v3: `https://v-iashin.github.io/detector`

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Perception and Scene Understanding

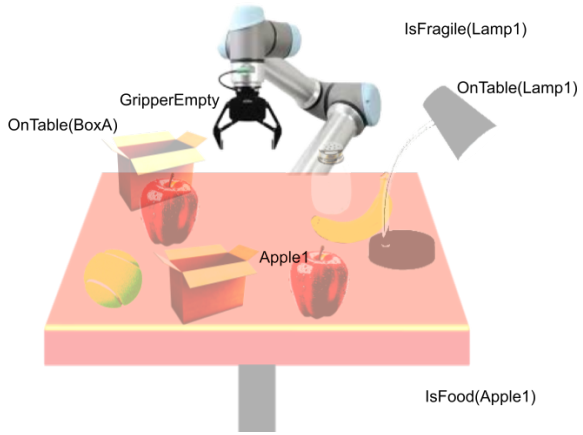
- **3D Localization:** Using the depth map, the robot computes the apple's 3D position in its camera coordinate frame.
- This point is transformed into world coordinates using the robot's calibration data.
- Uncertainty of estimation influences planning (e.g., grasp robustness).



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Task Interpretation - Task Planning

Translate the scene into a symbolic representation



The representation (e.g., using PDDL or STRIPS domain) is designed to be used by the high-level symbolic planner that finds a sequence of actions or policy to achieve the goal.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Task Interpretation - Task Planning

- A planner translate the goal condition (“pick up an apple”) into a concrete sequence of actions.
- Possible representation:

Goal: (holding(apple) = true)

State Features: apple_on(table), gripper_empty ...

Actions: move_to(apple), grasp(apple)...

Is this a sufficient representation for our setting ? What if we have more than one way to achieve the goal (here, more than one apple)?

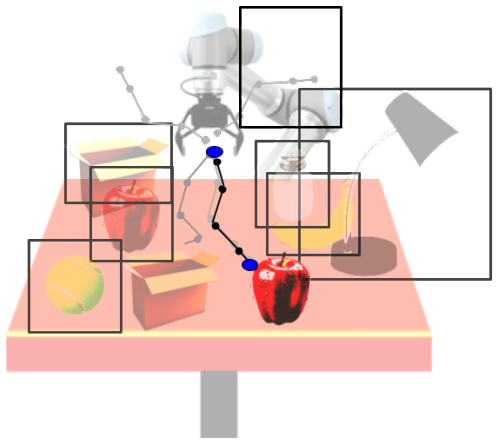
Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Task Interpretation - Motion Planning

A motion planner is used to compute a **collision-free trajectory** for the robot's arm to reach the apple.

The planner ensures:

- desired position is reached (the gripper's approach vector aligns with the apple)
- no collision occurs with the table or nearby objects.
- kinematic feasibility (joint limits, reachability, etc.) is satisfied.

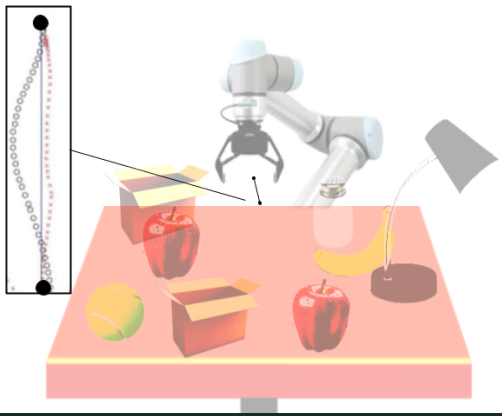


Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Control and Execution

”Move the arm, close the gripper, and lift the apple”

- **Trajectory Execution** - low-level controller (e.g., PID, MPC) uses feedback from joint encoders and force sensors to follow the planned trajectory.
- **Force control loop** - tactile sensors measure contact pressure so that the controller can regulate the grasp force.
- **Vision-based feedback** - (visual servoing) refines the alignment if necessary; the apple disappears from the table view and appears in gripper view.



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Uncertainty Management and Error Handling

- If the apple is occluded, the robot performs active exploration (moves head/camera).
- If the robot's gripper misses the apple, it may trigger a failure recovery behavior (e.g., retry with a new pose estimate).
- If grasp fails (slip detected or object dropped), the system may replan or retry.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Learning and Adaptation

The objective is to improve performance over time.

- **Reinforcement Learning:** The robot can learn grasp strategies through trial and error based on success rates.
- **Imitation Learning:** The robot can learn from demonstrations of humans picking apples.
- **Active Perception:** If uncertainty in pose estimation is high, the robot may reposition its camera to obtain a better view before acting.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Human–Robot Interaction

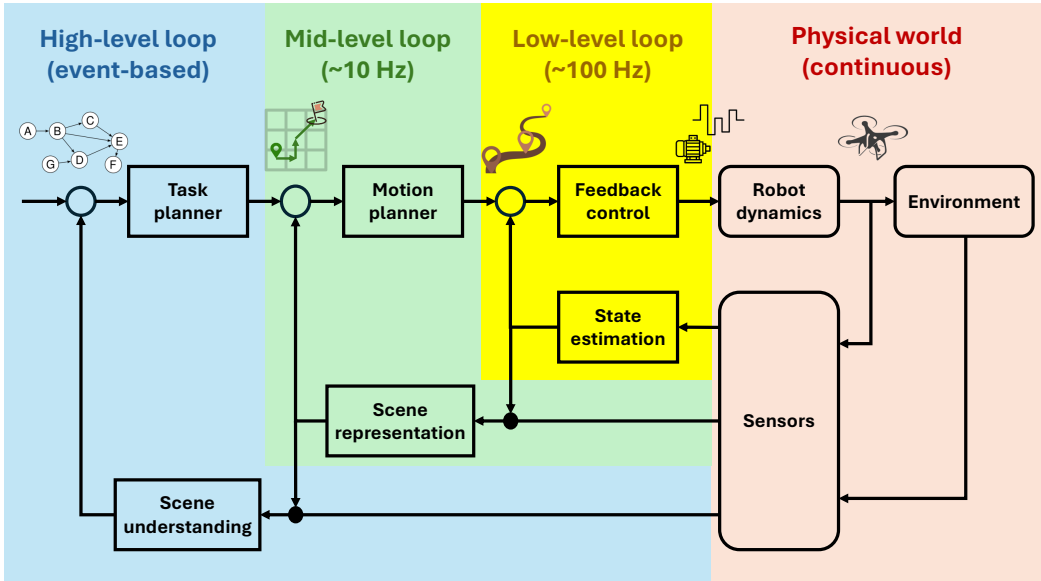
- If the robot works with a human:
- It may confirm the object by asking: “Should I pick up this apple?”
- Safety layers ensure compliant motion when humans are nearby (collision avoidance, dynamic replanning).

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Designing Complex systems

Perception–planning–control pipeline

Stage	Subsystem	Representative Methods
Perception	Vision & tactile sensing	CNNs, point-cloud segmentation, SLAM
Task Planning	Symbolic reasoning	PDDL, STRIPS, LLM-based planners
Motion Planning	Continuous optimization	RRT*, CHOMP, TrajOpt
Control	Dynamics and feedback	PID, impedance, MPC
Learning	Adaptive behavior	RL, imitation learning, active perception
HRI	Natural interaction	Speech, gesture, safety-aware motion



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Course Overview

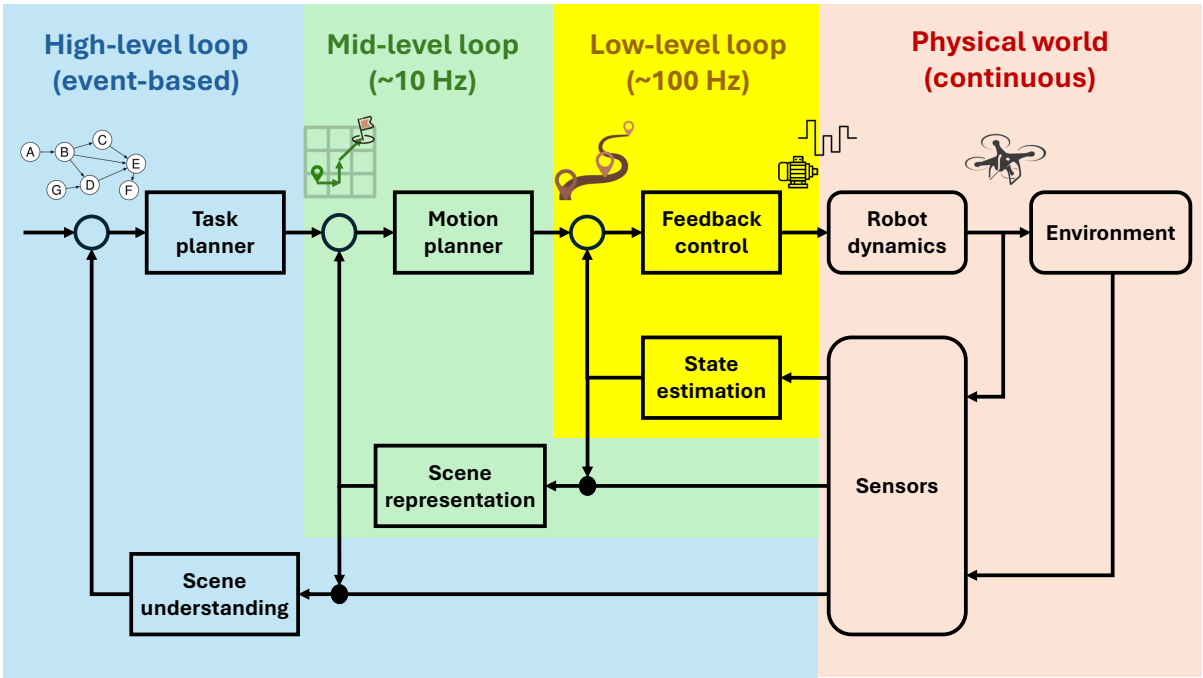
What is this class about ?

- From observations to decisions
- From low-level to high-level reasoning
- From single agent to multi-Agent
- From theory to practice to theory ...

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Our Focus

Integrating task planning, motion planning, and control.



Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Course plan and relationship to IAI

Week	Lecture Topic	IAI
1	Introduction Approaches to decision making in AI	Intro Reactive vs. cognitive
2	Components of a robotic agent and Three-Layered Decision Making in Robotics	NA NA
3	Task planning for robotics and Reactive control	Search
4	Motion planning	Search
5	Planning under partial observability and noisy sensing	MDP, POMDP, MCTS, RL
6	Planning under partial observability and noisy sensing	—
7	Integrated task and motion planning	—
8	Integrated task and motion planning (contd.)	—
9	ML and RL in robotics	ML, RL
10	Integrated learning and planning for robotic agents	—
11	Integrated learning and planning (contd.)	—
12	Multi-robot systems	Multi-agent Search
13	Conclusion	NA

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Schedule

- **Teaching:** Sarah Keren
  `sarahk@cs.technion.ac.il`
- **TA:** Guy Azran
  `guy.azran@campus.technion.ac.il`
- **Schedule:**
  - Lesson: Wednesday: 14:30-16:30
  - Lab / tutorial: Monday 16:30-17:30
  - Reception Hours: by request.
- **Communication with staff:**
  - Most communication should be done by the representative.

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Course Structure

- Weekly 2-hour class and a 1-hour tutorial.
- We will have 3 homework assignments and an exam.
- Quiz (MAGEN) week 8

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Useful Links

- 3rd Summer School on Cognitive Robotics (2019): https://sites.usc.edu/cognitive-robotics/schedule/
- TASP - Technion Autonomous Systems Program https://www.youtube.com/channel/UCaOlOHpGQc5Rza5bXzgLObw/featured
- MIT Robotics seminar: https://www.youtube.com/channel/UCK2tKzmSFFnpFhUXtRKjvnQ?app=desktop

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview

Robotics at the Technion

Figure 6: Some of the Technion's robotics researchers

Objectives

History of Robotics

Studying Robotics

CLAIR lab and Team

Our Running Example

Course Overview

Summary

- We motivated the study of robotics
- We highlighted the relationship between task planning, motion planning and control

What next ?

- Examine different AI Approaches and their relevance to robotics

Objectives
History of Robotics
Studying Robotics
CLAIR lab and Team
Our Running Example
Course Overview